

NIPAWIN, SK, Canada

WMO: 713120

Lat: 53.3331N

Lon: 104.0047W

Elev: 1220

StdP: 14.06

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-26.1	-38.6	0.6	-30.8	-33.9	0.8	-25.8	25.3	13.7	22.3	12.4	5.5	190	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	83.7	66.0	80.5	64.6	77.5	63.1	69.6	79.6	67.3	76.9	65.3	74.4	10.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	100.5	74.8	63.7	92.5	72.2	61.8	86.2	70.0	34.5	79.8	32.5	76.7	30.9	74.7	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.1	20.3	18.1	DB	-39.6	90.9	5.5	3.5	-43.5	93.5	-46.8	95.5	-49.9	97.5	-53.9	100.0	
			WB	-39.8	74.3	5.6	2.3	-43.8	76.0	-47.1	77.3	-50.3	78.7	-54.3	80.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.1	0.1	3.9	16.6	35.7	49.9	59.5	63.7	62.0	52.7	38.1	19.6	5.0
	DBStd	25.49	15.98	14.17	13.99	11.18	8.92	6.42	5.03	5.82	7.94	8.67	12.23	14.72
	HDD50	7185	1546	1292	1037	440	115	7	0	1	61	380	911	1395
	HDD65	11400	2011	1712	1502	879	472	186	84	127	372	834	1361	1860
	CDD50	1366	0	0	0	10	111	291	426	374	143	11	0	0
	CDD65	108	0	0	0	0	4	20	45	35	4	0	0	0
	CDH74	1542	0	0	0	6	117	275	488	511	142	3	0	0
	CDH80	333	0	0	0	1	25	58	85	131	33	0	0	0
Wind	WSAvg	8.7	8.4	8.3	8.6	9.4	10.2	8.9	7.7	7.8	8.9	9.0	8.7	8.3
Precipitation	PrecAvg	16.5	0.7	0.6	0.7	1.2	1.4	3.3	3.5	2.0	1.4	1.1	0.7	0.7
	PrecMax	22.6	1.6	1.1	2.3	3.3	3.7	6.7	6.4	4.8	3.3	3.2	1.7	1.7
	PrecMin	12.0	0.3	0.2	0.0	0.2	0.0	1.4	0.7	0.6	0.2	0.0	0.2	0.3
	PrecStd	3.1	0.4	0.2	0.6	0.7	0.9	1.3	1.7	1.1	1.0	0.8	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	38.3	47.1	72.0	83.0	86.4	86.5	89.9	84.8	70.4	49.8	39.1
		MCWB	34.9	33.9	39.6	52.6	59.1	65.8	70.3	67.7	63.5	55.4	42.4	34.9
	2%	DB	33.4	33.3	41.2	65.4	76.9	80.4	82.2	83.6	78.0	62.3	43.3	33.8
		MCWB	30.9	30.9	36.2	49.1	55.7	63.6	68.4	65.6	60.3	49.9	38.1	31.3
	5%	DB	28.3	29.0	38.1	58.8	72.6	76.8	79.3	79.8	72.8	56.4	38.6	28.8
		MCWB	26.6	27.1	34.4	46.2	53.8	61.2	66.8	64.7	57.8	47.2	35.2	27.0
	10%	DB	23.6	24.8	35.3	52.7	68.3	73.3	76.4	76.0	67.9	51.5	35.1	25.2
		MCWB	22.4	22.9	32.6	43.2	51.8	59.4	65.2	62.6	55.9	44.9	32.6	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.1	34.7	40.2	53.9	61.8	70.1	74.3	70.8	65.5	56.4	43.2	35.4
		MCDB	38.2	37.5	46.6	69.9	75.6	80.8	82.6	82.4	78.2	68.6	48.6	38.8
	2%	WB	31.1	30.8	36.6	50.4	58.3	66.3	70.7	68.3	62.3	51.4	38.6	31.2
		MCDB	33.3	33.1	40.5	62.5	72.8	76.3	79.6	79.5	75.7	58.7	42.6	33.7
	5%	WB	26.7	27.1	34.7	47.1	55.6	63.8	68.3	66.3	59.3	48.2	35.5	27.0
		MCDB	28.3	28.8	37.8	57.7	68.7	73.5	77.0	77.0	70.3	54.6	38.4	28.6
	10%	WB	22.5	22.8	32.8	43.7	53.2	61.6	66.3	64.2	56.6	45.4	32.7	23.8
		MCDB	23.8	24.4	35.3	52.3	66.2	70.1	74.4	73.3	66.4	51.2	35.0	25.0
Mean Daily Temperature Range	5% DB	MDBR	17.1	18.7	19.5	20.4	24.6	21.0	21.0	22.8	22.4	17.5	14.5	15.9
		MCDBR	20.0	18.2	18.2	31.2	32.1	27.6	26.1	30.1	32.0	27.3	18.7	18.0
	5% WB	MCWBR	18.1	16.6	14.2	17.6	15.4	13.7	14.9	14.9	16.6	16.7	14.2	16.2
		MCDWR	19.7	18.9	17.3	28.4	27.7	23.2	24.2	25.9	29.3	25.2	17.6	17.7
Clear-Sky Solar Irradiance	taub		0.247	0.255	0.281	0.329	0.344	0.373	0.373	0.368	0.316	0.287	0.247	0.248
		taud	2.274	2.305	2.335	2.393	2.407	2.363	2.370	2.389	2.520	2.550	2.469	2.238
	Ebn at Noon		238	275	290	287	287	278	276	271	275	263	244	210
		Edh at Noon	19	25	30	33	35	36	36	33	26	20	16	16
All-Sky Solar Radiation	RadAvg		315	659	1156	1553	1792	1797	1857	1543	1077	613	284	216
	RadStd		29	38	79	129	161	132	116	132	107	94	39	29

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page